GERMAN INTERNATIONAL UNIVERSITY (GIU) • FACULTY OF INFORMATICS

Practical Assignment: Mobile Home Screen Re-creation

Course: Information & Communication Architecture (ICA) • BINF 301

Core Goal: Hands-on HTML5 & CSS layout mastery from scratch in Notepad

Open Assignment Portal ↗ youneszied.github.io/ICA_lab_5_Assignment-/

GIU GERMAN INTERNATIONAL UNIVERSITY الجامعة الألمانية الدولية

How the deterministic assignment portal works

Each student receives a cryptographically unique assignment generated from their university Student ID.

1

Open Portal

Visit the public GitHub Pages generator portal to begin:

Launch Portal ↗

`youneszied.github.io/ICA_lab_5_Assignment-/`

2

Enter Student ID

Input your university ID (e.g. `19006583`) to seed the generator.

Deterministic Seed

3

Personalized Specs

Instantly computes your theme (Food, Bakery, Grocery, Pharmacy, Electronics) and brand palette.

5 Distinct Verticals

4

Download Package

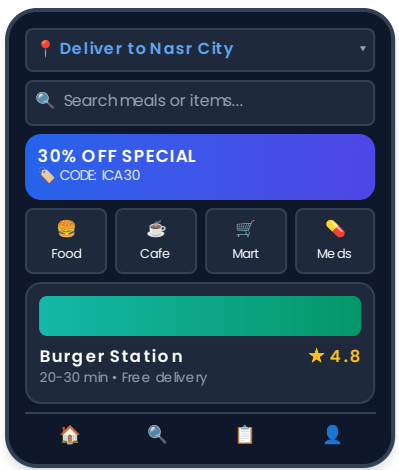
Download your individualized 3-page assignment brief (PDF) and target mockup (PNG).

3-Page Brief + Target PNG

CANVAS SECURITY ARCHITECTURE The target UI on the portal is rendered strictly onto an HTML5 `<canvas>` element—there are **zero HTML DOM elements** to inspect or copy. You must craft the semantic HTML and CSS structure yourself from your design brief.

Anatomy of the target mobile screen

Your submission must construct the following 6 core functional components inside a mobile wrapper:



1 Delivery Location Header

Location indicator with pin emoji (📍 **DELIVER TO ...**), current street/city text, and delivery selector dropdown.

2 Search Bar Component

Styled `<input type="search">` with search emoji prefix (🔍) and personalized placeholder text.

3 Promotional Hero Banner

Dual-gradient CSS container with `linear-gradient`, discount headline, subtext, and coupon badge (🏷️ **CODE: ...**).

4 Category Grid / Flexbox

Exactly 4 themed category tiles featuring assigned native emojis (🍔, ☕, etc.), rounded cards, and category titles.

5 Featured Store Card

CSS gradient cover banner, bold store name, rating badge (★ 4.8), delivery time tag, and minimum order metadata.

6 Fixed Bottom Navigation

Anchored bottom `<nav>` container with 4 interactive tab items: 🏠 Home, 🔍 Search, 📋 Orders, and 👤 Profile.

```
Mobile Viewport Constraint:  max-width: 400px; margin: 0 auto; min-height: 100vh;
```

</>

Step-by-step implementation guide (build in Notepad)

Follow this architectural roadmap to assemble your screen cleanly from blank file to final prototype:

Step 1: Viewport & Centered Container

Create `<StudentID>.html` in Notepad. Set the viewport meta tag and wrap content in container styles: `max-width: 400px; margin: auto;`

Step 2: Header & Search Input

Code the semantic `<header>` with delivery location label and an accessible `<input type="search">` using your assigned placeholder.

Step 3: Dual-Gradient Promo Hero

Build a hero card styled with `linear-gradient(135deg, Primary, Secondary)`, rounded corners, discount title, and coupon badge.

Step 4: Category Section (Flexbox or Grid)

Use `display: flex` or `display: grid; grid-template-columns: repeat(4, 1fr)` to lay out 4 category cards with native emojis and text labels.

Step 5: Featured Store Card

Build the featured store card with a CSS gradient cover banner, bold store name, rating pill badge (★ 4.8), and delivery time tag.

Step 6: Anchored Bottom Navigation

Anchor the `<nav>` bar to the bottom viewport with `position: fixed; bottom: 0; max-width: 400px;` and distribute the 4 tabs with flexbox.

Asset guidelines: Mandatory Unicode emoji policy

Automated headless grading evaluates your code in an isolated CI sandbox. Adhere strictly to these asset rules:

✗ Strictly Prohibited: External Images

- **No local machine paths:** Linking `C:\Users\...\icon.png` or `file:///...` fails immediately in automated testing.
- **No external image hosts:** Hotlinking external CDN URLs or image servers causes 404 network timeouts in headless testing.
- **No SVG files or icon fonts:** Do not load FontAwesome, Lucide scripts, or external sprite sheets.

Result: Unresolved links trigger automated 404 deduction

✓ Mandatory: Native Unicode Emojis

- **Text Emojis:** Place native Unicode emojis directly in your HTML text nodes: 📍 Location, 🔍 Search, 🍔 Food, ☕ Coffee, ★ Rating, 🏷️ Voucher, 🏠 Home, 📋 Orders, 👤 Profile.
- **CSS Visual Gradients:** Store cover banners and promo backgrounds must be rendered using pure CSS `linear-gradient(...)`.
- **Zero Dependencies:** 100% self-contained, lightning-fast rendering, and works offline anywhere.

Result: 100% reliable headless test rendering

How to type emojis in Notepad: Windows: Press `Win + .` • Mac: Press `Cmd + Ctrl + Space`

Official grading rubric breakdown (100 points)

Your submission is scored out of 100 points across three automated evaluation pillars:

Structure & Semantics — 40 Pts

- Centered wrapper: `max-width: 400px;` and `margin: auto;`
- Semantic tags: `<header>`, `<main>`, `<section>`, `<nav>`
- Exact element counts: 4 categories, 1 featured store card, 4 bottom nav items
- Proper input element: `<input type="search">`

DOM Architecture & Hierarchy

Layout & Styling — 40 Pts

- Modern alignment: `display: flex` or `grid` on categories and navigation
- Fixed bottom navigation anchored to the viewport bottom
- Color accuracy: Assigned brand primary and secondary palette applied
- Touch ergonomics: Padding, border-radii, and clean contrast hierarchy

Computed CSS & Visual Fidelity

Academic Integrity — 20 Pts

- Pure human-written code matching your unique Student ID seed
- Automated detection scans for prompt injection honeypot tokens
- Inspection for AI artifacts (markdown code fences, default font stacks)
- Submissions matching another student's seed receive 0 points

Seed Consistency & AI Defense

Scored automatically via Headless Chromium DOM & CSS evaluation

Total Available: 100 Points

Submission instructions & final checklist

Ensure strict compliance with file naming and technical constraints before uploading:

Strict File Naming Rule

`<StudentID>.html`

Example: For student ID **19006583**, name your file strictly: `19006583.html`

Invalid names like `index.html` or `assignment.html` are rejected by the automated ingestion pipeline.

Portal: youneszied.github.io/ICA_lab_5_Assignment-/ ↗

Submit: Microsoft Forms link & QR on next slide

Pre-Submission Verification Checklist

- [✓] **Single self-contained file:** All CSS in `<style>` in `<head>`
- [✓] **Mobile container bounds:** `max-width: 400px; margin: auto;`
- [✓] **Asset rule:** All icons are native emojis (zero external images)
- [✓] **Zero JavaScript:** No `<script>` tags or JS event handlers
- [✓] **Seed consistency:** Matches parameters from your assigned Student ID

Deadline: Exactly 1 Week from your lab session • Submit via Microsoft Forms

ASSIGNMENT 1 SUBMISSION PORTAL



Open Microsoft Forms Submission ↗

https://forms.cloud.microsoft/Pages/ResponsePage.aspx?id=twQ5o3wq-k-WjR0V5Byu8yfIedfI9QBOq86huAYW4xZURURPNVlNN1lHVDA3U1FaSllTWVJBOE4yTC4u